CO2AT2

Name: Dharshavardhan S.B
Reg: 192420039

①. Cryptography for IoT Devices :

A). Constraints of IoT Devices :

- o) Limited processing power
- o) Low battery capacity
- o) Limited memory and storage
- o) Restricted network bandwidth

B). Recommendation : ECC

ECC is proffered over RSA because it provides Strong Security with Smaller keys, Low memory storage usage, generally less power consumption. it is suitable for resource constrainted IOT devices.

Conclusion : ECC is more effecient for secure communicati in Smarter home IOT Systems.

⑤. Secure Data transmission using Block Cipher

A). Why does this issue occur?

ECB mode encrypts identical plaintext blocks into identical ciphertext block using the same key this reveals data pattern & reduces Confidentiallity.

B). Reccommendation mode : CBC

Cipher Block chaining (CBC) uses an initialization Vector to make identical plaintext blocks produce differenta ciphertext blocks.

Justification: CBC hides Repeated patterns better than ECB when implemented with a secure unpredictable IV. it also requires padding for incomplete blocks.

Conclusion: CBC is more secure than ECB for protecting repeated financial data pattern